



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Faculty of Science and Technology (FST)
Department of Computer Science (CS)
Undergraduate Program

COURSE PLAN	SEMESTER: FALL 2024-2025
I. Course Code and Title CSC 2108 Introduction to Database II. Credit 3 credit hours (1-hour 40 minutes theory and 2-hours 40 minutes Lab per week) III. Nature Core Course for CSE IV. Prerequisite: CSC 1205 Object Oriented Programming 1	V. Vision: Our vision is to be the preeminent Department of Computer Science through creating recognized professionals who will provide innovative solutions by leveraging contemporary research methods and development techniques of computing that is in line with the national and global context. VI. Mission: The mission of the Department of Computer Science of AIUB is to educate students in a student-centric dynamic learning environment; to provide advanced facilities for conducting innovative research and development to meet the challenges of the modern era of computing, and to motivate them towards a life-long learning process.

VII - Course Description

- Identify the drawbacks of file-based management system and the necessity of Database management system; Use of modern tools used in Database management system.
- Understanding different types of terminologies used in Database management system.
- Discuss different tools and techniques for better performance of Database management system.
- Execute necessary and sufficient SQLs; Design ER Models and Diagrams; Use of different types of Normalization process; Analyze a system with a view to DBMS implementation; Understand different types of joining and use of different complex queries.

VIII - Course outcomes (CO) Matrix

By the end of this course, students should be able to:

COs*	CO Description	Level of Domain***			PO Assessed***
		C	P	A	
CO1	Adapt to data models, database systems, the integration and use of relational systems and relational languages, query processing and the theory of RDBMS.		4		PO-e-2
CO2**	Simplify the fundamental concepts underlying database systems and illustration of ER diagram and case study with tools like Oracle 10g and MySQL	4			PO-c-2
CO3	Construct the database schema and create ER diagram with tools as vision, diagram, lucid chart etc.		4		PO-e-2
CO4**	Create DML, DDL using Oracle and connection with OJDBC/JDBC for existing JAVA application		4		PO-e-2

C: Cognitive; P: Psychomotor; A: Affective Domain

* CO assessment method and rubric of COs assessment is provided in later section

** COs will be mapped with the Program Outcomes (POs) for PO attainment

*** The numbers under the 'Level of Domain' columns represent the level of Bloom's Taxonomy each CO corresponds to.

**** The numbers under 'PO Assessed' column represent the POs each CO corresponds to.

IX - Topics to be covered in the class and/or lab *

Time Frame	CO Mapped	Topics	Teaching Activities	Assessment Strategy(s)
Week 1	CO1, CO4	OBE, Mission & Vision of AIUB. Introduction To Database Systems, Oracle Environment Setup, Schema Setup, Database Terminology, Basic Concept of Database	Lecture, Question-Answer, Lab Practice	Quiz, Term Exam, Lab Work, Assignment
Week 2	CO1, CO4	Writing Basic SQL Statements, SQL Sorting and Restricting	Lecture, Question-Answer, Lab Practice	Quiz, Term Exam, Lab Work, Assignment
Week 3	CO2, CO4	ER Diagrams, Single-Row Functions	Lecture, Question-Answer, Lab Practice	Quiz, Term Exam, Lab Work, Assignment
Week 4	CO1, CO2, CO4	Aggregate Function, Entity Relationship Modeling with case studies	Lecture, Question-Answer, Lab Practice	Quiz, Term Exam, Lab Work, Assignment
Week 5	CO3	Details Lecture on DDL, DML	Lecture, Question-Answer, Lab Practice	Quiz, Term Exam, Lab Work, Assignment
Week 6	CO4	Data Constraints Details	Lecture, Question-Answer, Lab Practice	Quiz, Term Exam, Lab Work, Assignment
Mid Term (Week 7)				
Week 8	CO1, CO2, CO3	Functional Dependencies and related keys discussion, Normalization and its necessities.	Lecture, Question-Answer, Lab Practice	Quiz, Lab Work, Assignment, Project
Week 9	CO1, CO2, CO3	Discussion on First Normal Form and Second Normal Form	Lecture, Question-Answer, Lab Practice	Quiz, Lab Work, Assignment, Project
Week 10	CO1, CO2, CO3, CO4	Third Normal Form and Joining Details	Lecture, Question-Answer, Lab Practice	Quiz, Lab Work, Assignment, Project
Week 11	CO3, CO4	View, User Access Control, Project discussion	Lecture, Question-Answer, Lab Practice	Quiz, Lab Work, Assignment, Project
Week 12	CO1, CO2, CO3	Schema Diagram, Relational Algebra	Lecture, Question-Answer, Lab Practice	Quiz, Lab Work, Assignment, Project
Week 13	CO4	Project discussion and review syllabus	Lecture, Question-Answer, Lab Practice	Quiz, Lab Work, Assignment, Project
Final Term (Week 14)				
Project Submission/Defense/Viva (Week 15)				

* The faculty reserves the right to change, amend, add, or delete any of the contents.

X - Mapping of PO to Courses and K, P, A

PO Indicator ID	PO Indicators Definition (As per the requirement of Wks)	Domain	K	P	A
PO-c-2	Develop process for complex computer science and engineering problems considering cultural and societal factors.	Cognitive Level 4 (Analyzing)	K5	P1 P3 P7	
PO-e-2	Use modern engineering and IT tools for prediction and modeling of complex computer science and engineering problems considering the practice in computer science and engineering discipline.	Psychomotor Level 4 (Articulation)		P1 P4 P5	

XI – K, P, A Definitions

Indicator	Title	Description
K5	Engineering Design (Technology)	Knowledge that supports engineering design in a practice area.
P1	Depth of knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach
P3	Depth of Analysis Required	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models.
P4	Familiarity of issues	Involve infrequently encountered issues
P5	Extent of applicable codes	Are outside problems encompassed by standards and codes of practice for professional engineering
P7	Interdependence	Are high level problems including many component parts or sub-problems.

XII – Mapping of CO Assessment Method and Rubric

The mapping between Course Outcome(s) (COs) and The Selected Assessment method(s) and the mapping between Assessment method(s) and Evaluation Rubric(s) is shown below:

COs	Description	Mapped POs	Assessment Method	Assessment Rubric
CO1	Adapt to data models, database systems, the integration and use of relational systems and relational languages, query processing and the theory of RDBMS.	PO-e-2	Project	Rubric for Project
CO2	Simplify the fundamental concepts underlying database systems and illustration of ER diagram and case study with tools like Oracle 10g and MySQL	PO-c-2	Project Report	Rubric for Report
CO3	Construct the database schema and create ER diagram with tools as vision, diagram, lucid chart etc.	PO-e-2	Project application verification	Rubric for Report

CO4	Creating DML, DDL using Oracle and connection with OJDBC/JDBC for existing JAVA application	PO-e-2	Project	Rubric for Project
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XIII – Evaluation and Assessment Criteria

CO1: Adapt to data models, database systems, the integration and use of relational systems and relational languages, query processing and the theory of RDBMS.

Assessment Attribute/Criteria	Missing/Incorrect (0)	Inadequate (1)	Satisfactory (2)	Excellent (3)
Content knowledge	Relational Database System is not discussed with example	Relational Database System is discussed but did not provide any example	Relational Database System is discussed but provided example with very few entity	Relational Database System is discussed but provided example with sufficient entity
Case Study	Did not discuss the case study	Discussed the case study but could not identify all the entities and attributes	Discussed the case study, identified all the entities but missed few relationships	Discussed the case study, identified all the entities, relationships and attributes

CO2: Simplify the fundamental concepts underlying database systems and illustration of ER diagram and case study with tools like Oracle 10g and MySQL

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
ERD Design	Design ERD based on the described case study				
Correctness	Designed ERD supports database design concept				

CO3: Construct the database schema and create ER diagram with tools as visio, dia diagram, lucidchart etc.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Problem Analysis	Background information of the problem and definition of key solution.				
Proposed Solution	Provide a solution that incorporates valid Database Management System.				
Normalization	Normalization should be done Up to 3NF				
Schema	Proper use of DDL and DML				

CO4: Creating DML, DDL using Oracle and connection with OJDBC/JDBC for existing JAVA application

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Query questions	Develop effective query questions				
Query analysis	Correctness and execution of the queries				
Submission	Submission of Project should be submitted within deadline following the minimum project requirements with a Project report.				

Teamwork and VIVA	Teamwork and VIVA Contribution of each team member and it's justification through a VIVA session.
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XIV- Course Requirements

- Students are expected to attend at least 80% class.
- Students are expected to participate actively in the class.
- For both terms, there will be at least 2 quizzes based on the theoretical knowledge and conceptual understanding of the topic covered discussed in the classes.
- Submit report based on the given course related problems.
- Submission of assignment and projects should be in due time.

XV – Evaluation & Grading System*

The following grading system will be strictly followed in this class

MID TERM		FINAL TERM	
Attendance	10%	Attendance	10%
Quiz	30%	Quiz	30%
Lab performance/Assignment	10%	Lab performance/Assignment	10%
Midterm written exam	50%	Project	50%
Total	100%	Total	100%
Grand Total 100% = 40% of Midterm + 60% of Final Term			

Letter	Grade Point	Numerical %
A+	4.00	90-100
A	3.75	85 - < 90
B+	3.50	80 - < 85
B	3.25	75 - < 80
C+	3.00	70 - < 75
C	2.75	65 - < 70
D+	2.50	60 - < 65
D	2.25	50 - < 60
F	0.00	< 50
I		Incomplete
W		Withdrawal
UW		Unofficially Withdrawal

* The evaluation system will be strictly followed as par the AIUB grading policy.

* CO attainment will be achieved with 60% of the evaluation marks.

XVI – Textbook/ References

1. Modern Database Management (Sixth Edition) by Fred R. McFadden, Jeffrey A. Hoffer, Mary B. Prescott
2. Database System Concepts (Fifth Edition) by Henry F. Korth, S. Sudarshan, A. Silberschatz
3. Oracle-database-10g-sql-fundamentals-1-student-guide-volume-1

4. SQL and Relational Theory: How to Write Accurate SQL Code by C.J. Date
5. Database Systems: A Practical Approach to Design, Implementation and Management (4th Edition) by Thomas M. Connolly, Carolyn E. Begg
6. Fundamentals of Database Systems, 5th Edition by Ramez Elmasri, Shamkant B. Navathe
7. Database Design and Relational Theory: Normal Forms and All That Jazz by C. J. Date
8. An Introduction to Database Systems 8th Edition, by C.J. Date

XVII - List of Faculties Teaching the Course

FACULTY NAME	SIGNATURE
1. METHILA FARZANA WOISHE	
2. AKIB AHMED	
3. ANIK KUMAR SAHA	
4. DR. RAJARSHI ROY CHOWDHURY	
5. JUBAYER AHAMED	
6. JUENA AHMED NOSHIN	
7. MD SAJID BIN-FAISAL	
8. MD. MAHMUDUR RAHMAN	
9. NYME AHMED	
10. RAZUAN KARIM	
11. SAIKAT BAUL	
12. SIFAT RAHMAN AHONA	
13. SYED NAFIUL SHEFAT	
14. TAMANNA ZAMAN BRISTY	

XVI – Verification

Prepared by:  Methila Farzana Woishe <i>Course Convener</i> Date: October 25, 2024	Moderated by: Dr. M. Mahmudul Hasan <i>Point Of Contact</i> <i>OBE Implementation Committee</i> Date:.....	Checked by: Dr. Akinul Islam Jony <i>Head (Undergraduate Program)</i> <i>Department of Computer Science</i> Date:.....
Verified by: Dr. Md. Abdullah-Al-Jubair <i>Director</i>	Certified by: Prof. Dr. Dip Nandi <i>Associate Dean,</i>	Approved by: Mr. Mashioor Rahman <i>Dean,</i>

<i>Faculty of Science & Information Technology</i> Date:.....	<i>Faculty of Science & Information Technology</i> Date:.....	<i>Faculty of Science & Information Technology</i> Date:.....
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